

浙大量子 2020

一、(1) $E_n = -\frac{e^2}{2an^2}$ 简并度 $2n^2$

(2) 精细结构形成有两个原因: 相对论修正和自旋-轨道耦合.

超精细结构和电子与质子之间的偶极矩之间的磁相互作用有关

(3) $-\frac{\hbar^2}{2m^*} \nabla^2 \psi + V^* \psi = E \psi$

$V^* = \frac{1}{\epsilon_r} V, m^* = 1.2m$

$a = 0.07a_0$

$E_n = -\frac{Z^2 e^2}{2an^2}$

$= -\frac{1.2^2 \left(\frac{e}{2}\right)^2}{2 \times 0.07 a_0 n^2} = -\frac{1.2^2}{0.07 \times 9} \frac{e^2}{2a_0 n^2}$
 $\approx -2.3 \frac{e^2}{2a_0 n^2}$

二、(1) $\psi(a^+) = \psi(a^-)$

$\psi(-a^+) = \psi(-a^-)$

$\psi'(a^+) - \psi'(a^-) = \frac{2mA}{\hbar^2} \psi(a^+)$

$\psi'(-a^+) - \psi'(-a^-) = \frac{2mA}{\hbar^2} \psi(-a^+)$

(2) 偶宇称 $\psi(x) = \begin{cases} Be^{k_1 x}, & x \leq -a \\ \cosh k_1 x, & -a < x < a \\ Be^{-k_1 x}, & x \geq a \end{cases}$

由连续性条件

$Be^{-k_1 a} = \cosh k_1 a \quad (1)$

$-\cosh k_1 a - k_1 \sinh k_1 a - k_1 Be^{-k_1 a} = \frac{2mA}{\hbar^2} Be^{-k_1 a} \quad (2)$

由 $\cosh^2 k_1 a - \sinh^2 k_1 a = 1$ 得:

$B^2 = -\frac{\hbar^4 k_1^2 e^{-2k_1 a}}{(2mA)^2 + 4MA\hbar^2 k_1}$

同理, $C^2 = \frac{\hbar^4 k_2^2 e^{-2k_2 a}}{(2mA)^2 + 4MA\hbar^2 k_2}$

将(2)代入(1)中, 有

$-k_1 (\sinh k_1 a + \cosh k_1 a) = \frac{2mA}{\hbar^2} Be^{-k_1 a}$

$-k_1 e^{k_1 a} = \frac{2mA}{\hbar^2} Be^{-k_1 a}$

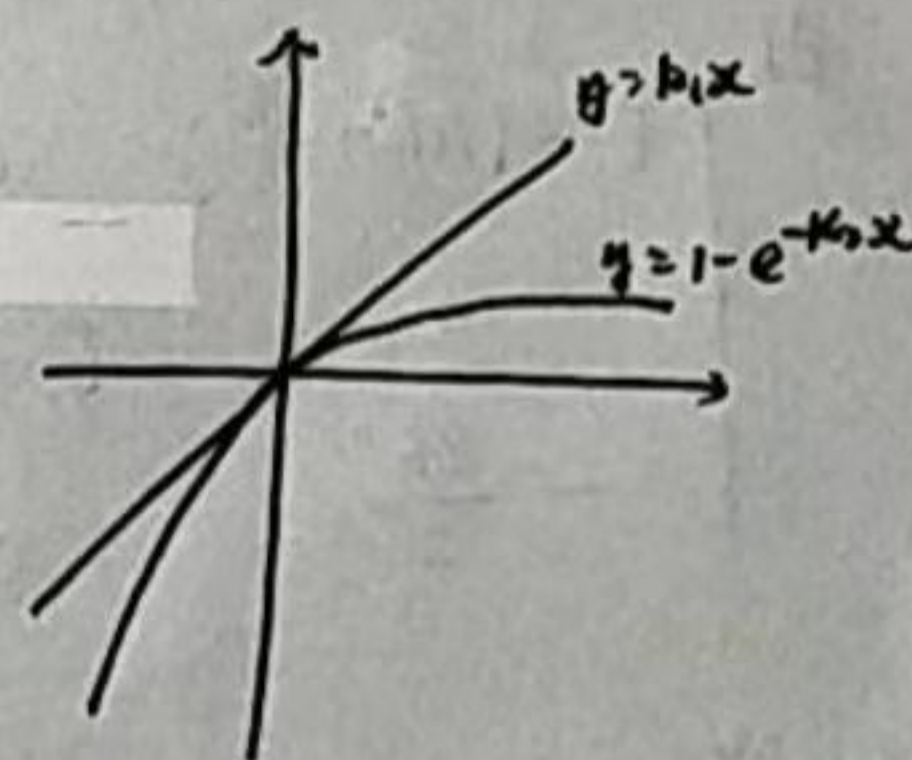
$B = -\frac{\hbar^2}{2mA} k_1 e^{2k_1 a}$

同理 $C = \frac{\hbar^2}{2mA} k_2 e^{k_2 a}$

(3) 无题

(4) $y_1 = k_1 x$

$y_2 = 1 - e^{-k_2 x}$



在零点, $y_2' = k_2$

当 $k_1 = k_2$ 时,

仅有一个解

$k_1 \neq k_2$ 时, 有两个解

三、(1) 自旋为1. 在 S^2, S_z 表象中.

$$\hat{S}_x = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad \hat{S}_z = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{H} = A\hat{S}_x + B\hat{S}_z^2$$

$$= A\hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} + \frac{B\hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} A\hbar + \frac{B\hbar^2}{2} & 0 & \frac{B\hbar^2}{2} \\ 0 & B\hbar^2 & 0 \\ \frac{B\hbar^2}{2} & 0 & -A\hbar + \frac{B\hbar^2}{2} \end{pmatrix}$$

$$\text{由} \begin{vmatrix} E - (A\hbar + \frac{B\hbar^2}{2}) & 0 & -\frac{B\hbar^2}{2} \\ 0 & E - B\hbar^2 & 0 \\ -\frac{B\hbar^2}{2} & 0 & E + (A\hbar - \frac{B\hbar^2}{2}) \end{vmatrix} = 0$$

$$\text{得: } (E - B\hbar^2) \left[(E - \frac{B\hbar^2}{2})^2 - (A\hbar)^2 - (\frac{B\hbar^2}{2})^2 \right] = 0$$

$$E_1 = B\hbar^2$$

$$E_2 = \frac{B\hbar^2}{2} + \frac{\sqrt{A^2 + B^2}}{2} \hbar$$

$$E_3 = \frac{B\hbar^2}{2} - \frac{\sqrt{A^2 + B^2}}{2} \hbar$$

$$(2). \text{初始时 } |\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

本征值的本征态.

$$\text{令 } A\hbar = a, \quad \frac{B\hbar^2}{2} = b, \quad \sqrt{a^2 + b^2} = c$$

$$E_1 = 2b, \quad \psi_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$E_2 = b + c, \quad \psi_2 = \frac{1}{\sqrt{b^2 + (a+c)^2}} \begin{pmatrix} c+a \\ 0 \\ b \end{pmatrix}$$

$$E_3 = b - c, \quad \psi_3 = \frac{1}{\sqrt{b^2 + (a-c)^2}} \begin{pmatrix} a-c \\ 0 \\ b \end{pmatrix}$$

$$|\psi(t)\rangle = c_1 e^{-iE_1 t/\hbar} |\psi_1\rangle + c_2 e^{-iE_2 t/\hbar} |\psi_2\rangle + c_3 e^{-iE_3 t/\hbar} |\psi_3\rangle$$

$$c_1 = \langle \psi_1 | \psi(0) \rangle = 0$$

$$c_2 = \langle \psi_2 | \psi(0) \rangle = \frac{a+c}{\sqrt{b^2 + (a+c)^2}}$$

$$c_3 = \langle \psi_3 | \psi(0) \rangle = \frac{a-c}{\sqrt{b^2 + (a-c)^2}}$$

$$\hat{S}_z \text{ 本征值为 } -\hbar \text{ 本征态为 } \psi = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

处于 $-\hbar$ 概率

$$P = |\langle \psi | \psi(t) \rangle|^2$$

$$= \left| c_1 e^{-iE_1 t/\hbar} \frac{b}{\sqrt{b^2 + (a+c)^2}} + c_2 e^{-iE_2 t/\hbar} \frac{b}{\sqrt{b^2 + (a-c)^2}} \right|^2$$

$$= \left| \frac{b(a+c)}{b^2 + (a+c)^2} e^{-i(b+c)t/\hbar} + \frac{b(a-c)}{b^2 + (a-c)^2} e^{-i(b-c)t/\hbar} \right|^2$$

$$= \frac{a-c}{b} + \frac{b^2}{2c^2} \cos \frac{2ct}{\hbar}$$

四. (1) 由归一化得

$$\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = \int_{-\infty}^{+\infty} |A|^2 e^{-\alpha^2 x^2} dx$$

$$= 2|A|^2 \int_0^{\infty} e^{-\alpha^2 x^2} dx = 2|A|^2 \cdot \frac{1}{2} \sqrt{\frac{\pi}{\alpha^2}} = 1$$

$$A = \left(\frac{\alpha}{\pi}\right)^{1/4}$$

概率分布函数 $P = |\psi(x)|^2 = \sqrt{\frac{\alpha}{\pi}} e^{-\alpha^2 x^2}$

$$(2) \langle x \rangle = \int_{-\infty}^{+\infty} \psi^*(x) x \psi(x) dx$$

$$= \int_{-\infty}^{+\infty} |A|^2 e^{-\alpha^2 x^2} x dx = \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} e^{-\alpha^2 x^2} x dx = 0$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \psi^*(x) x^2 \psi(x) dx$$

$$= \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} x^2 e^{-\alpha^2 x^2} dx = -\sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} \frac{x}{2\alpha^2} d e^{-\alpha^2 x^2}$$

$$= \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} e^{-\alpha^2 x^2} \frac{1}{2\alpha^2} dx = \frac{1}{2\alpha^2}$$

$$\langle p \rangle = \int_{-\infty}^{+\infty} \psi^*(x) \left(-i\hbar \frac{d}{dx}\right) \psi(x) dx$$

$$= \int_{-\infty}^{+\infty} \sqrt{\frac{\alpha}{\pi}} e^{-\alpha^2 x^2/2} (-i\hbar) (-\alpha^2 x) e^{-\alpha^2 x^2/2} dx$$

$$= i\hbar \alpha^2 \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} e^{-\alpha^2 x^2} x dx = 0$$

$$\langle p^2 \rangle = \int_{-\infty}^{+\infty} \psi^*(x) \left(-\hbar^2 \frac{d^2}{dx^2}\right) \psi(x) dx$$

$$= -\hbar^2 \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{+\infty} e^{-\alpha^2 x^2} (\alpha^4 x^2 - \alpha^2) dx$$

$$= \frac{\alpha^2 \hbar^2}{2}$$

$$(3) \text{ 由 } -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \frac{1}{2} m \omega^2 x^2 \psi(x) = E \psi(x)$$

$$\Rightarrow E = \frac{\hbar^2 \alpha^2}{2m}$$

$$\Rightarrow \psi(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\frac{mE x^2}{\hbar^2}}$$

$$(4) \langle \Delta x^2 \rangle = \langle (x - \bar{x})^2 \rangle = \langle x^2 - 2x\bar{x} + \bar{x}^2 \rangle$$

$$= \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\alpha^2}$$

$$\langle \Delta p^2 \rangle = \langle (p - \bar{p})^2 \rangle = \langle p^2 - 2p\bar{p} + \bar{p}^2 \rangle$$

$$= \langle p^2 \rangle - \langle p \rangle^2 = \frac{\alpha^2 \hbar^2}{2}$$

$$\langle \Delta x^2 \rangle \langle \Delta p^2 \rangle = \frac{\hbar^2}{4} \geq \frac{\hbar^2}{4}$$

五. 一维谐振子

$$\hat{H} = \frac{1}{2m} \hat{p}^2 + \frac{1}{2} m \omega^2 x^2$$

$$\hat{a}^{\dagger} = \frac{1}{\sqrt{2m\hbar\omega}} (i\hat{p} + m\omega x)$$

$$\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}} (-i\hat{p} + m\omega x)$$

$$\Rightarrow \begin{cases} \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}} (\hat{a}^{\dagger} - \hat{a}) \\ x = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}^{\dagger} + \hat{a}) \end{cases}$$

$$\text{六. (1)} \quad E_1^{(1)} = H'_{11} = \langle \psi_1 | H' | \psi_1 \rangle$$

$$E_1^{(2)} = \frac{|H'_{21}|^2}{E_1^{(0)} - E_2^{(0)}} = \frac{|\langle \psi_2 | H' | \psi_1 \rangle|^2}{E_1^{(0)} - E_2^{(0)}}$$

= 二级近似能量

$$E_1 = E_1^{(0)} + \langle \psi_1 | H' | \psi_1 \rangle + \frac{|\langle \psi_2 | H' | \psi_1 \rangle|^2}{E_1^{(0)} - E_2^{(0)}}$$

$$|\psi\rangle = |\psi_1^{(0)}\rangle + \frac{\langle \psi_2 | H' | \psi_1 \rangle}{E_1^{(0)} - E_2^{(0)}} |\psi_2^{(0)}\rangle$$

E_2 同理.

$$(2) \quad E_1 = E_2, \text{ 二重简并. } |\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$H' = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}.$$

$$W_{11} = \langle \psi_1 | H' | \psi_1 \rangle = \alpha$$

$$W_{12} = \langle \psi_1 | H' | \psi_2 \rangle = 0$$

$$W_{21} = 0$$

$$W_{22} = \langle \psi_2 | H' | \psi_2 \rangle = \alpha.$$

$$W = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}.$$

$$\text{由 } \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = E^{(1)} \begin{pmatrix} a \\ b \end{pmatrix}.$$

$$\Rightarrow E^{(1)} = \alpha, |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\psi_1\rangle$$

$$E^{(1)} = -\alpha, |\psi\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |\psi_2\rangle$$